

## CHARITON, IA, USA

WMO: 725469

Lat: 41.019N

Lon: 93.359W

Elev: 320

StdP: 97.54

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04913

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -19.2      | -16.9 | -23.8                         | 0.5 | -19.2 | -21.3 | 0.6 | -15.4 | 12.8                  | -5.6 | 11.9 | -6.1 | 3.6                   | 310  | 0.533 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.5                   | 33.8            | 25.4 | 32.4 | 24.9 | 31.1 | 24.2 | 26.9                | 32.0 | 26.0 | 30.7 | 25.1 | 29.6 | 4.5                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.8                            | 22.0 | 30.8 | 24.7 | 20.6 | 29.6 | 23.5 | 19.0 | 28.3 | 87.2          | 32.1 | 82.1 | 31.0 | 78.1 | 29.4 | 29.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 11.1          | 9.7  | 8.6 | DB                         | -23.7 | 36.2               | 4.3 | 1.9                                                | -26.8 | 37.6       | -29.4 | 38.7       | -31.8 | 39.7       | -35.0 | 41.1 | (4) |
| (5)               |      |     | WB                         | -24.2 | 28.2               | 3.7 | 0.9                                                | -26.8 | 28.9       | -29.0 | 29.4       | -31.0 | 29.9       | -33.7 | 30.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.0        | -3.7  | -1.9  | 4.3   | 11.2  | 17.0  | 22.2  | 24.7  | 23.1  | 19.2  | 12.2  | 5.1   | -1.7  | (6)  |  |
|                                                                        | DBStd       | 11.28       | 6.86  | 6.93  | 6.41  | 5.34  | 4.53  | 3.55  | 3.24  | 3.05  | 4.46  | 5.32  | 5.93  | 6.44  | (7)  |  |
|                                                                        | HDD10.0     | 1579        | 426   | 336   | 199   | 47    | 3     | 0     | 0     | 0     | 1     | 38    | 166   | 363   | (8)  |  |
|                                                                        | HDD18.3     | 3260        | 684   | 567   | 437   | 219   | 81    | 8     | 1     | 2     | 42    | 202   | 396   | 621   | (9)  |  |
|                                                                        | CDD10.0     | 1957        | 0     | 2     | 22    | 84    | 219   | 366   | 455   | 404   | 278   | 105   | 21    | 2     | (10) |  |
|                                                                        | CDD18.3     | 596         | 0     | 0     | 1     | 6     | 39    | 124   | 198   | 148   | 69    | 11    | 1     | 0     | (11) |  |
|                                                                        | CDH23.3     | 5450        | 0     | 0     | 6     | 73    | 329   | 1117  | 1971  | 1261  | 603   | 87    | 3     | 0     | (12) |  |
|                                                                        | CDH26.7     | 1988        | 0     | 0     | 1     | 14    | 84    | 399   | 827   | 447   | 200   | 17    | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 4.1         | 4.5   | 4.5   | 4.8   | 5.0   | 4.3   | 3.6   | 3.0   | 2.8   | 3.4   | 4.0   | 4.5   | 4.4   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 917         | 26    | 33    | 52    | 98    | 125   | 138   | 109   | 104   | 89    | 61    | 48    | 34    | (15) |  |
|                                                                        | PrecMax     | 1344        | 57    | 62    | 137   | 204   | 243   | 329   | 315   | 253   | 184   | 154   | 124   | 120   | (16) |  |
|                                                                        | PrecMin     | 594         | 14    | 6     | 1     | 40    | 30    | 25    | 26    | 3     | 42    | 8     | 1     | 2     | (17) |  |
|                                                                        | PrecStd     | 184         | 9     | 15    | 34    | 38    | 52    | 73    | 69    | 67    | 41    | 39    | 30    | 25    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 14.0  | 17.9  | 23.7  | 28.7  | 32.0  | 33.9  | 37.2  | 35.0  | 33.0  | 28.9  | 22.9  | 16.1  | (19) |  |
|                                                                        |             | MCWB        | 8.7   | 11.8  | 16.8  | 19.9  | 24.0  | 25.7  | 25.1  | 24.8  | 23.7  | 20.8  | 15.7  | 12.6  | (20) |  |
|                                                                        | 2%          | DB          | 11.0  | 13.9  | 20.2  | 25.3  | 28.9  | 32.4  | 34.7  | 32.8  | 31.2  | 26.0  | 19.9  | 12.4  | (21) |  |
|                                                                        |             | MCWB        | 7.1   | 9.6   | 14.1  | 16.7  | 21.6  | 24.6  | 26.2  | 25.3  | 23.2  | 18.3  | 14.1  | 9.0   | (22) |  |
|                                                                        | 5%          | DB          | 7.7   | 10.9  | 17.5  | 22.7  | 27.1  | 31.0  | 32.7  | 31.1  | 28.9  | 23.0  | 17.1  | 9.7   | (23) |  |
|                                                                        |             | MCWB        | 4.9   | 7.4   | 12.6  | 15.6  | 19.9  | 23.9  | 25.5  | 24.7  | 22.5  | 16.8  | 12.0  | 7.4   | (24) |  |
|                                                                        | 10%         | DB          | 5.0   | 7.4   | 14.1  | 20.2  | 24.9  | 29.0  | 31.2  | 28.9  | 27.2  | 21.0  | 14.0  | 7.1   | (25) |  |
|                                                                        |             | MCWB        | 2.6   | 4.6   | 9.7   | 14.1  | 18.8  | 22.9  | 24.9  | 23.6  | 21.1  | 16.0  | 9.5   | 4.6   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 10.2  | 13.9  | 17.9  | 20.7  | 24.7  | 26.9  | 28.7  | 28.0  | 25.8  | 22.8  | 16.9  | 13.6  | (27) |  |
|                                                                        |             | MCDB        | 12.7  | 16.5  | 22.5  | 26.9  | 29.9  | 32.5  | 34.2  | 32.7  | 30.5  | 27.2  | 21.6  | 16.0  | (28) |  |
|                                                                        | 2%          | WB          | 7.2   | 9.9   | 15.5  | 18.3  | 22.7  | 25.8  | 27.4  | 26.7  | 24.6  | 20.2  | 14.8  | 10.0  | (29) |  |
|                                                                        |             | MCDB        | 9.6   | 13.5  | 18.7  | 23.8  | 27.6  | 30.9  | 32.9  | 31.2  | 29.5  | 24.3  | 18.5  | 11.3  | (30) |  |
|                                                                        | 5%          | WB          | 5.0   | 7.3   | 13.1  | 16.5  | 21.2  | 24.7  | 26.3  | 25.5  | 23.5  | 18.4  | 12.6  | 6.9   | (31) |  |
|                                                                        |             | MCDB        | 8.0   | 10.2  | 17.1  | 21.6  | 25.5  | 29.4  | 31.5  | 29.6  | 27.6  | 21.8  | 16.5  | 9.1   | (32) |  |
|                                                                        | 10%         | WB          | 2.8   | 4.8   | 10.1  | 14.9  | 19.7  | 23.6  | 25.3  | 24.3  | 22.1  | 16.3  | 10.2  | 4.4   | (33) |  |
|                                                                        |             | MCDB        | 5.2   | 7.5   | 14.3  | 19.5  | 23.6  | 27.9  | 29.8  | 28.0  | 26.1  | 20.1  | 13.8  | 6.7   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.9   | 9.1   | 10.1  | 11.2  | 10.5  | 10.3  | 10.5  | 10.3  | 11.6  | 11.0  | 9.9   | 8.4   | (35) |  |
|                                                                        |             | MCDBR       | 12.3  | 13.4  | 14.7  | 14.9  | 12.5  | 11.6  | 11.8  | 12.1  | 13.0  | 13.3  | 13.7  | 11.3  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 9.3   | 9.2   | 9.2   | 9.0   | 6.9   | 6.2   | 5.9   | 6.5   | 6.6   | 7.6   | 8.9   | 8.6   | (37) |  |
|                                                                        |             | MCWBR       | 11.9  | 12.4  | 12.7  | 13.0  | 11.0  | 10.5  | 10.6  | 10.4  | 11.3  | 11.1  | 12.4  | 10.5  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.287 | 0.317 | 0.332 | 0.387 | 0.408 | 0.418 | 0.432 | 0.420 | 0.379 | 0.338 | 0.298 | 0.286 | (40) |  |
|                                                                        |             | taud        | 2.433 | 2.348 | 2.397 | 2.259 | 2.268 | 2.282 | 2.237 | 2.292 | 2.354 | 2.467 | 2.514 | 2.457 | (41) |  |
|                                                                        | Ebn at Noon |             | 875   | 892   | 919   | 884   | 870   | 859   | 842   | 843   | 856   | 859   | 858   | 846   | (42) |  |
|                                                                        |             | Edh at Noon | 78    | 99    | 106   | 131   | 133   | 132   | 137   | 126   | 110   | 88    | 72    | 71    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.94  | 2.90  | 3.77  | 4.78  | 5.62  | 6.25  | 6.29  | 5.47  | 4.66  | 3.13  | 2.14  | 1.65  | (44) |  |
|                                                                        | RadStd      |             | 0.19  | 0.21  | 0.32  | 0.35  | 0.45  | 0.60  | 0.47  | 0.33  | 0.33  | 0.36  | 0.18  | 0.16  | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (45 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page